

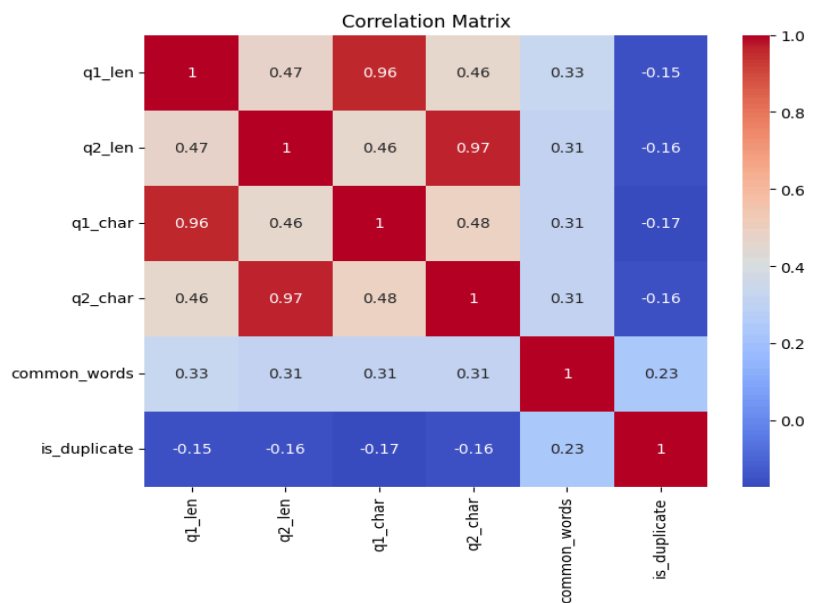
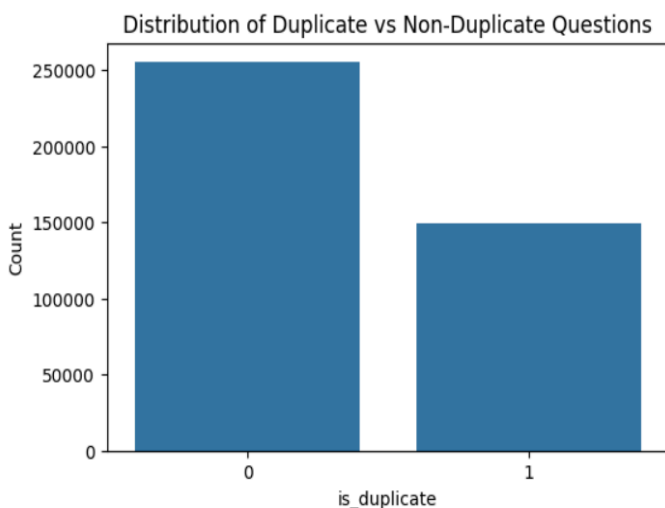
Task Assignment

1. Introduction

This project aims to analyze question pairs to predict whether two questions are duplicates. The primary goal is to explore patterns in question text and evaluate the performance of different models for classification. The project addresses challenges such as class imbalance, text preprocessing and semantic similarity. It compares approaches including Logistic Regression and Siamese LSTM networks.

2. Exploratory Data Analysis (EDA)

- **Dataset size:** 404,319 rows $\times$ 6 columns.
- **Features:** 4 numerical (id, qid1, qid2, is_duplicate), 2 categorical (question1, question2).
- **Missing values:** question1 (1), question2 (2)
- **Class distribution:** Non-duplicate $\approx$ 63%, Duplicate $\approx$ 37% (slight imbalance).
- **Text analysis:** Variability in word and character length of questions; common phrases identified.
- **Outliers & inconsistencies:** 4 problematic rows found and removed.
- **Correlations:** Features like question length, character count and common words analyzed for relationship with is_duplicate.



3. Text Preprocessing

- **Drop Missing Rows:** Removed rows where question1 or question2 was missing to ensure clean data.
- **Tokenization:** Split questions into individual words for analysis.
- **Lowercasing:** Standardized all text to lowercase to avoid duplicate tokens.
- **Remove Special Characters & Punctuation:** Cleaned text to retain only meaningful words.
- **Stopword Removal:** Removed common stopwords and tokenized cleaned text.
- **Lemmatization:** Normalized words to their root forms for consistency.
- **Feature Tuning:** Optimized max_features and ngram_range for TF-IDF vectorization.
- **Word Embeddings:** Created Word2Vec embeddings by averaging token vectors to represent question semantics numerically.

4. Model Create and Training

Baseline – Logistic Regression

- **Features:** TF-IDF vectors of question1 and question2 were used. Absolute differences and element-wise multiplication were combined to form the final feature set.
- **Target:** is_duplicate.
- **Handling Class Imbalance:** class_weight='balanced' was used in LogisticRegression to account for unequal class distribution.
- **Training:** Data was split into 80% training and 20% testing, and the Logistic Regression model was trained on the training set.

Siamese Shared LSTM

- **Text Representation:** Tokenized questions were padded to a maximum sequence length of 40. Word embeddings of dimension 100 were used.
- **Model Architecture:** A shared Bidirectional LSTM encoder was used to extract features from both questions. Outputs were merged using concatenation of absolute difference, element-wise multiplication, and original encoded vectors, followed by fully connected dense layers with dropout.

- **Handling Class Imbalance:** Class weights were computed from the training labels and applied during model training.
- **Training:** The model was trained with early stopping on validation loss, batch size 512, for 5 epochs.

5. Model Evaluation

The performance of both models was assessed using standard classification metrics, including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Logistic Regression served as a strong baseline, achieving solid overall performance. After further tuning and experimentation, the Siamese LSTM model demonstrated superior results across all key metrics, confirming its effectiveness for the task.

Logistic Regression Results

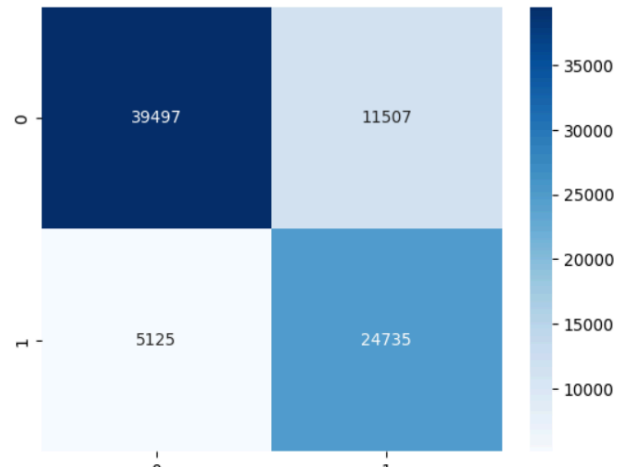
Accuracy: 0.7943

Precision: 0.6825

Recall: 0.8284

F1-Score: 0.7484

ROC-AUC: 0.8848



Siamese LSTM Results

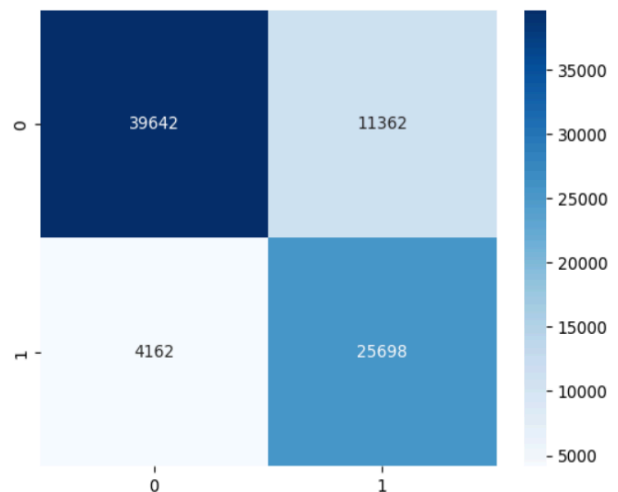
Accuracy: 0.8080

Precision: 0.6934

Recall: 0.8606

F1-Score: 0.7680

ROC-AUC: 0.8971



6. Model Tuning and Hyperparameter Optimization

Both Logistic Regression and Siamese LSTM were optimized for better performance. Class weights handled imbalance, dropout and early stopping prevented overfitting, and different architectures, activations, and optimizers were tested. The final configuration improved sensitivity to duplicate question pairs.

7. Conclusion

After evaluating both models, the Siamese LSTM, following tuning and optimization, demonstrated superior performance compared to the Logistic Regression baseline. Its higher accuracy, F1-score, and ROC-AUC indicate that it provides a more effective and reliable solution for the task.

